Hugo BOULENC

Curriculum Vitae

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GitHub Page *LinkedIn*

Professional Experience

2026 – present **Freelance in Scientific Machine Learning & Data Science**, *Freelance*, Montpellier, France.
I help companies and research teams turn complex scientific models into fast, actionable tools. I focus on two key bottlenecks: speeding up numerical simulations from hours to seconds using surrogate models and calibrating physical models with thousands of parameters automatically from data.

2020 – 2021 **Consulting Aerospace Engineer**, *CNES / ALCADIA*, Paris & Toulouse, France.
Modelization of Cryogenics Systems for the reusable launcher demonstrator Themis.

2020 **Internship in Fluid Dynamics and Machine Learning**, *ArianeGroup*, Les Mureaux, France.
Development of a surrogate model based on CFD calculations and Machine Learning for Ariane 6.

Education

2022 – 2026 **PhD in Applied Mathematics**, *INSA Toulouse / Institut de Mathématiques de Toulouse*, Toulouse, France.
PhD in Applied Mathematics on Physics-Informed Machine Learning methods applied to hydraulics and hydrology models for floods simulation. Supervised by J. Monnier (INSA Toulouse/IMT), P.-A. Garambois (INRAE Aix-en-Provence) and R. Bouclier (INSA Toulouse/ICA/IUF) at INSA Toulouse.

2019 – 2020 **Specialization Degree in Mechanics, Aeronautics and Aerospace Engineering**, *Centrale-Supélec*, Gif-sur-Yvette, France.
Final year of the "Ingénieur Centralien" curriculum, advanced courses in Fluid Dynamics.

2015 – 2020 **Msc in General Engineering, major in Aeronautics and Aerospace**, *EPF École d'Ingénieur-e-s*, Montpellier & Sceaux, France.
Exchange Semester with Polytechnique Montréal in Fluid Dynamics & Heat Transfer Science.

Teaching Experience

2022 – 2024 **Teaching Assistant**, *INSA Toulouse*, Toulouse, France.
Tutorial and Practical Sessions in Data Assimilation, PDE and Fourier Analysis for 2^{nd} to 5^{th} year students.

Publications

2025 **H. Boulenc**, R. Bouclier, P.-A. Garambois and J. Monnier (2025). Spatially-distributed parameter identification by physics-informed neural networks illustrated on the 2D shallow-water equations. *Inverse Problems.*.

Conferences & Workshops

2025 **DTE & AICOMAS**, Paris, France.
Mini-Symposium presentation.

2024 **ECCOMAS**, Lisboa, Portugal.
Mini-Symposium presentation.

2024 **ANITI Days**, Toulouse, France.
Poster co-presentation with M. Allabou.

2024, 2025 **ANITI Chair PILearnWater Plenary Session**, Toulouse, France.

2022 – 2024 **ANR MUFFINS Plenary Sessions**, Aix-en-Provence & Toulouse, France.